Lab: Feature Engineering – Mind the Gap!

What to do with gaps in data sets?

- five step exercise

Mind_the_Gap_I

Mind_the_Gap_II

Mind_the_Gap_III

Mind_the_Gap_IV

Mind_the_Gap_V

- each step:
  - you will **work in groups in breakout rooms**
  - after time is up: return to the **main room** and **discuss the solution**
  - you will **get the sample solution** and **compare it to your own solution**
  - material for **next step** will be pushed to **277A repo**
  - repeat: next step

- You don’t need to turn anything in. You will have the complete solution at the end.

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What to do with gaps in data sets?

preparation:

- You will modify **`lists`** and **`dataframes`**. Both types **are mutable**! In order to keep the original data for each step use **`.copy`**

```
new_df = old_df.copy()
```

- use random number generators for sampling, eg. **`np.random.normal`**

- we will use **UMAP** (more details see Module 8), a **visualization tool** that projects **high dimensional data to two or three dimensions**. We will use UMAP for visualizing our data set unlike PCA, **UMAP does not remove dimensions**.

```
pip install umap-learn

import umap.umap_ as umap

newXY = umap.UMAP().fit_transform(X)
```

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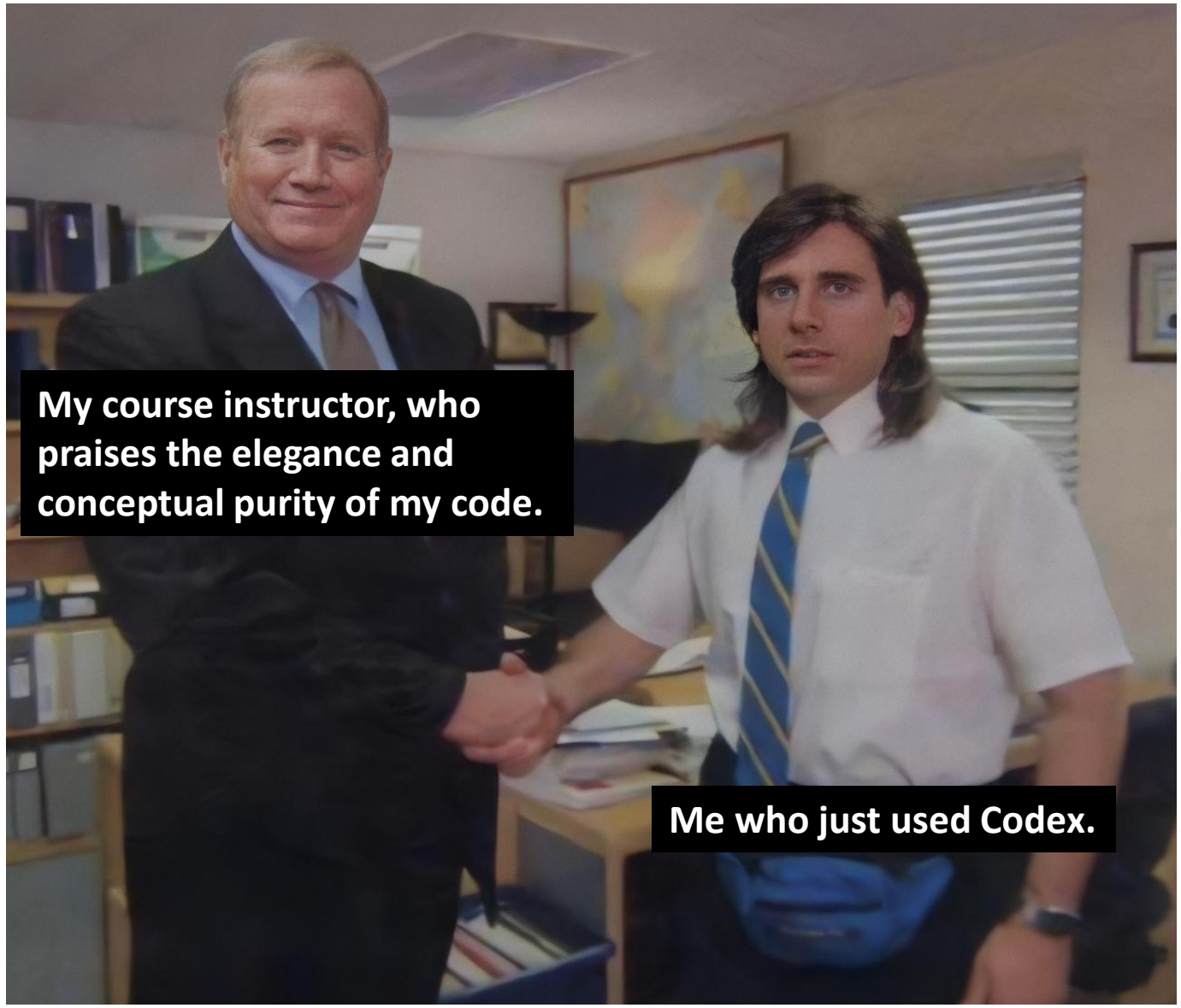
What to do with gaps in data sets? **Enjoy :)**